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Purpose/Policy:

This standard operating procedure is for the “TRIpack Sleever” to assist with setup, operation, changeover, cleaning, and troubleshooting.

Scope:

The purpose of the “TRIpack Sleever” is to apply a sleeved label onto cans as they pass through the sleever. This unit can sleeve 12oz, 16oz, and 19.2oz cans.

The “TRIpack Sleever” is comprised of two main pieces of equipment. The first being the sleever unit itself and the second being a steam tunnel. The sleever unit feeds labels from a spool onto cans as the cans pass by. The steam tunnel then shrinks the sleeves down to fit onto the can using heat. Before starting up or working on the “TRIpack Sleever” ensure that the E-Stop is pressed.

Definitions/Acronyms:

- SOP – Standard Operating Procedure
- PPE – Personal Protective Equipment
- HMI – Human Machine Interface (the touch screen used to control a machine)
- E-Stop – Emergency Stop button
- UHMW – Ultra-High Molecular Weight plastic (used for guides and wear strips)
- TRIpack Sleever – Machine that applies a sleeved label onto cans
- Steam Tunnel – Tunnel that uses heat to shrink sleeves onto cans after the sleever

Authorities/Responsibilities:

This SOP can be completed by an entry level packaging operator after proper training from a senior member of the Can line packaging team. The objective of this SOP is to gain the knowledge needed to operate the “TRIpack Sleever” and trouble shoot any issues that may arise.

Resources and Materials:

Operations:

- Correct sleeves for current packaging run
- Steam from the boiler
- Exacto knife
- Roll of colored tape
- 10mm wrench

Login Information

- Username: oper
- Password: oper

Changeovers:

- 4mm hex key
- 12oz Size Guide
- 16oz Size Guide
- 19.2oz Size Guide
- Login information

Basic Cleaning:

1. Isopropyl and rag for cleaning rollers.

Deep Clean:

2. Refer to SOP - Can line Zone 2 Cleaning
   2.1 N:\Operations\Production\Boston\Packaging\Canning Line\Canning 2025\Zone Cleaning 2025\Can Line Upper Level\Zone 2 - Sleever Area

Troubleshooting:

3. No Equipment required.

PPE Required:

1. Black heat resistant gloves for steam tunnel jams.

Procedures:

Setting up the TRIpack Sleever:

1. Prio to start up perform the following checks.
   1.1 Make sure the worm wheel is inline and set to the correct distance.
      1.1.1 Loosen orange handles before trying to adjust with wheel.
   1.2 Inspect the machine for damaged or missing parts.
   1.3 Make sure all guards are on and the doors are closed.
   1.4 Make sure all electrical enclosures are closed.
   1.5 Roll the film feed and supplication wheels with your hand to ensure they rotate properly.
   1.6 Make sure there are no tools or foreign objects located in the sleever.
   1.7 Make sure all E-Stop buttons are working properly
   1.8 Make sure the main power disconnect is on.
   1.9 Determine the can size can you will be running.
   1.10 Use the changeover section of the SOP if you require a size change.

1.11 You can now begin the setting up of the sleever.

2. Using the loading area table remove one half of the core plates.
   - 2.1 Slide the roll of film into the middle of the core plate.
   - 2.2 Reattach the other side of the film holder so that white plastic part of the core plate is secure inside the core of the film roll.
   - 2.3 Place the film reel back onto the film carriage in the correct orientation.
3. Once you have the TRIpack set up to the correct size you will start by threading film roll.
   - 3.1 To thread the film, raise the pinch roller, thread the film over the film assist roll. Reference schematic guide on the side of the roll carriage.
   - 3.2 Thread the film down into the fan box and under the guide rod.
   - 3.3 Thread the film through guide located at the top rear of the air box.
   - 3.4 Thread film over the dead plate and pass it under the guide bar.
   - 3.5 Pass the film under the brush bar and guide the film over the dead plate.
   - 3.6 Thread the film down towards the mandrel.
   - 3.7 Thread film between the guide rollers and over the plow portion of the mandrel.
   - 3.8 Pull down some extra length of film.
   - 3.9 Slide film over the main portion of the mandrel until the leading edge comes into contact with the side tension wheels.
   - 3.10 Turn the side tension wheels by hand to advance the film until the leading edge of the film contacts the front tension wheels. If the film is crinkling loosen wheels feed film through.
   - 3.11 Turn the front tension wheels by hand to advance the film until the leading edge contacts the feed wheels. If the film is crinkling loosen wheels feed film through.
   - 3.12 Using the detent knob, back the film wheels away from the mandrel.
   - 3.13 Slide film over the remaining portion of the mandrel until leading edge is close to the product conveyor.
   - 3.14 Use the detent knob to bring the feed wheels back into position against the mandrel. Continue turning the knob until a noticeable slipping is felt in the internal clutch.
   - 3.15 From the HMI Log in
   - 3.16 Use the “One Cycle” function on the HMI to cut the film until it is the correct length.
4. Starting the Seam tunnel
   - 4.1 Slowly turn the inlet ball valve on top of the unit to the open position to bring steam into the machine.
   - 4.2 On the HMI turn the exhaust tunnel on.
   - 4.3 Open the manifold doors (4 corners) to access internal controls.
   - 4.4 Slightly open the drainage manifold (bottom lever) to allow condensation to drain.
   - 4.5 Close the drainage manifolds.
   - 4.6 Close manifold doors.

Operation of the TRIpack Sleever:

1. To start the sleever follow the following steps.
    - 1.1 Access the run screen on the HMI.
    - 1.2 Press Tamp Enable
    - 1.3 Press Conveyor Start
    - 1.4 Press Infeed Screw Start
    - 1.5 Press the Machine Start button
    - 1.6 Inspect product for quality, let it run through the tunnels and see if the sleeves are shrinking correctly.
2. Once a roll of film is almost empty you should get a low film warning.
    - 2.1 If you require more sleeves to finish the run follow step 2 in the “Setting up TRIpack Sleever: section to have your next roll of sleeves ready to go.
    - 2.2 Once the first roll finishes make sure to stop the Sleever before the film is fed up through the pinch roller.
    - 2.3 Remove the empty film real assembly.
    - 2.4 Cut about ¼ off of the last label on the old roll.
    - 2.5 Cut about a ¼ of the new rolls first label.
    - 2.6 Insert the old film into the new one and line up the labels as best as you can.
    - 2.7 Make sure to tape the labels together.
    - 2.8 Start the Sleever back up.
    - 2.9 Watch for the sleeve that has been spliced and pull it off the line.

Changeover Guide:

1. Start by logging in.
    - 1.1 Username: oper
    - 1.2 Password: oper
2. Select size change screen on the HMI
    - 2.1 Enter the corresponding number for the correct size.
3. Next adjust the head height counter.
    - 3.1 Turn wheel (labeled 3) to the corresponding value listed on the set-up sheet.
4. Adjust the application vertical wheel counter.
    - 4.1 Turn wheel (labeled 1) to the corresponding value listed on the set-up sheet.
5. We will then swap out the size guide on the mandrel.
    - 5.1 Do this by loosening the 4mm hex bolts on the mandrel located underneath the blade assembly on the UHMW size guide.
    - 5.2 Remove the old size guide.
    - 5.3 Replace it with the corresponding size guide for the run size.
    - 5.4 Tighten down the 4mm hex bolts that were previously loosened.

6. Adjust “Application fail” photo eye so that it is below the mandrel.
    6.1 This will require using a 4mm hex key to loosen the rail.
    6.2 Once moved, tighten back down the rail.

Shutting Down TRIpack Sleever:

1. Stopping the sleever.
    1.1 At the end of the run press stop on the TRIpack sleever to stop operations.
2. Turning off the Steam Tunnel.
    2.1 Close the main steam valve on top of the steam tunnel.
    2.2 Wait for steam to stop coming from the machine.
    2.3 Then turn off sleever conveyor.
    2.4 Allow the surfaces to cool.
    2.5 Turn off tunnel exhaust on the HMI.
    2.6 Open manifold door to air out.
    2.7 Open all four-drainage manifold completely to allow for condensation to drain.
    2.8 Once all drained close drainage manifolds and doors.
3. Press E-Stop on HMI.

Basic Cleaning of the TRIpack Sleever:

1. Remove and bits of left over sleeves that might be stuck on the mandrel or in the blade assembly.
    1.1 Use caution when cleaning blade assembly.
2. Use a rag and some isopropyl to clean rollers and mandrel that might have residue leftover from sleeves.
    2.1 Spray rag with isopropyl, do not spray any surfaces in the sleever unit directly with isopropyl.
3. When there is no steam going to the steam tunnel open doors and look for any bad cans or labels that might be in the tunnel.

Deep Cleaning

Outline the deep cleaning process for the SOP equipment. See below for example:

1. Refer to SOP - Can line Zone 2 Cleaning
    1.1 N:\Operations\Production\Boston\Packaging\Canning Line\Canning 2025\Zone Cleaning 2025\Can Line Upper Level\Zone 2 - Sleever Area

Forms:

Not Applicable

Reference Documents:

Not Applicable

Common Issues:

1. Clearing HMI Alarm Stops
    - 1.1 Check the message on the HMI to determine the reason for the stop.
    - 1.2 If necessary, access the problem by opening the guard door.
    - 1.3 Close the guard door if opened.
    - 1.4 Reset machine by pressing reset, waiting until it's ready, then press run.
2. Application Fail
    - 2.1 Observe the conveyor and sensor for and obstructions.
    - 2.2 If any obstructions are found, remove them by accessing the area through the guard door.
    - 2.3 Close the guard door.
    - 2.4 Reset the machine.
    - 2.5 Use the "Cycle One" function to make sure sleeves are cutting the correct length.
3. Film Jam
    - 3.1 Observe mandrel for wrinkled or torn film.
    - 3.2 Observe recently sleeved product for ragged edges on the film.
    - 3.3 If film shows signs of damage, open the guard door to access mandrel.
    - 3.4 Slide film upwards until clear of mandrel.
    - 3.5 Remove any damaged film from the machine.
    - 3.6 Place undamaged film back on the mandrel.
    - 3.7 Manually advance film by turning the applicating wheels.
    - 3.8 Turn the knife assembly one complete revolution by hand.
    - 3.9 Remove the cut portion of film.
    - 3.10 Close the guard doors.
    - 3.11 Reset the machine.
    - 3.12 Use the "Cycle One" function to make sure sleeves are cutting the correct length.
    - 3.13 Start the machine.
4. Clearing Excessive Film Feed.
    - 4.1 Observe fan box for damaged film.
    - 4.2 Remove any damaged film if found.
    - 4.3 If no damaged film is found, the eye inside the fan box is not getting flagged every few seconds.
    - 4.4 Rethread the film as described in step 3 of "Setting up the TRIpack Sleever"
    - 4.5 Close guard door and reset the sleever.

4.6 Use the “Cycle One” function to make sure sleeves are cutting the correct length.

4.7 Start the machine.

5. Broken Film
   5.1 Repeat step 4 in this section.
6. Low Film
   6.1 This occurs as a warning when you are running low on film, keep an eye on the film and replace it when you are close to running out.
7. Can Jam in Steam Tunnel
   7.1 Using proper PPE (heat resistant gloves), open the doors of the steam tunnel and remove the jam. Close the doors when ready to start operations.
      7.1.1 Make sure to remove any bad cans or sleeves before starting up.

Document History and Approval:

Revise document history as described in the table and key below.

(Reference to following document: B1C.SOP.P001: Document Control and Approval Procedure Template)

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Approved by: Kyle Szymanski

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